

State Space Models (SSM)

Mamba-2 (Tri Dao)

$$h'(t) = A h(t) + B x(t)$$

$$y(t) = C h(t)$$

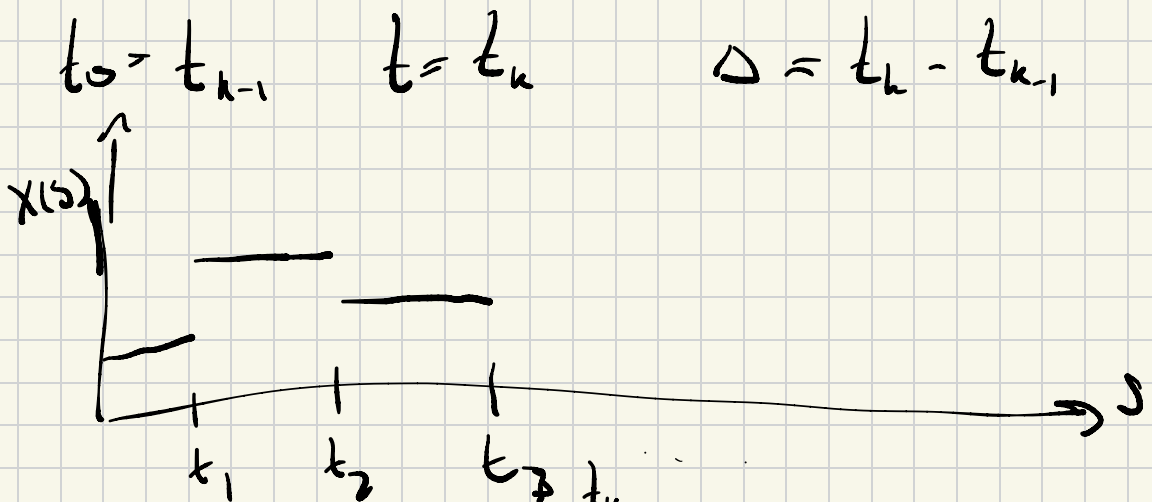


$$e^{-At} h'(t) - e^{-At} A h(t) = e^{-At} B x(t)$$

$$\frac{d}{dt} (e^{-At} h(t)) = e^{-At} B x(t)$$

$$e^{-At} h(t) - e^{-At_0} h(t_0) = \int_{t_0}^t e^{-As} B x(s) ds$$

$$h(t) - e^{A(t-t_0)} h(t_0) = \int_{t_0}^t e^{A(t-s)} B x(s) ds$$



$$h_k - e^{A\Delta} h_{k-1} = \int_{t_{k-1}}^{t_k} e^{A(t_k-s)} B x(s) ds$$

$$h_k = e^{A\Delta} h_{k-1} + \left(\int_{t_{k-1}}^{t_k} e^{A(t_k-s)} ds \right) B x_k$$

$$\int_{t_{k-1}}^{t_k} e^{A(t_k-s)} ds = \left| \tau = t_k - s \right| = \int_0^{\Delta} e^{A\tau} d\tau =$$

$$= A^{-1} (e^{A\Delta} - I)$$

$$h_k = e^{DA} h_{k-1} + \underbrace{A^{-1}(e^{DA} - I)B}_{\frac{1}{B}} x_k$$

$$y_k = C h_k$$

$$\begin{cases} h_k = \bar{A} h_{k-1} + \bar{B} x_k, & h_0 = 0 \\ y_k = C h_k \end{cases}$$

$$h_1 = \bar{B} x_1$$

$$h_2 = \bar{A} \bar{B} x_1 + \bar{B} x_2$$

$$h_3 = \bar{A} \bar{A} \bar{B} x_1 + \bar{A} \bar{B} x_2 + \bar{B} x_3$$

$$(\bar{A}^L \bar{B}, \bar{A}^{L-1} \bar{B}, \dots, \bar{B}) * (x_1, \dots, x_L) =$$

$$= (h_1, \dots, h_L)$$

$$h_k = \bar{A}_k h_{k-1} + \bar{B}_k x_k$$

$$y_k = C_k h_k$$

$$h_1 = \bar{B}_1 x_1$$

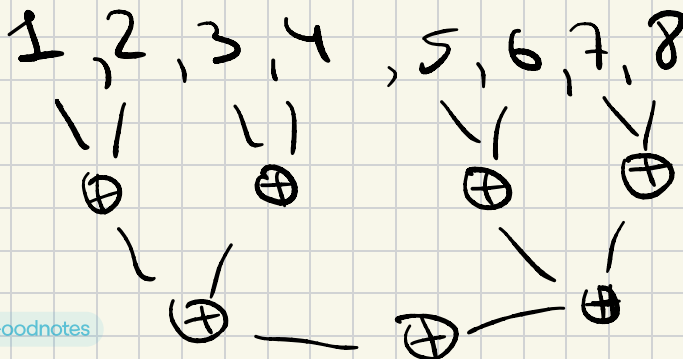
$$h_2 = \bar{A}_2 \bar{B}_1 x_1 + \bar{B}_2 x_2$$

$$h_3 = \bar{A}_3 \bar{A}_2 \bar{B}_1 x_1 + \bar{A}_3 \bar{B}_2 x_2 + \bar{B}_3 x_3$$

$$(a_1, b_1) \oplus (a_2, b_2) = (a_1, a_2, b_1 + a_2 b_2)$$

$$(\bar{A}_1, \bar{B}_1 x_1) \oplus (\bar{A}_2, \bar{B}_2 x_2) \oplus \dots \oplus (\bar{A}_L, \bar{B}_L x_L)$$

$$= (\bar{A}_1 \dots \bar{A}_L, h_L)$$



$L \leq \text{num-work}$
 $O(\log L)$

`jax.lax.associative_scan`